

DAY-4

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Python [conda env:base] Anaconda Toolbox

```
[5]: import pandas as pd
df=pd.read_excel("Employee-Management-Sample-Data.xlsx") #Importing excel dataset
df.head(10) #displays first 10 data from the dataset
```

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6
0	NaN	Excel Sample Data	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	Employee Management Data	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	Employee ID	Full Name	Department	Designation	Hire Date	Annual Salary (USD)
5	NaN	S1001	John Smith	Human Resources	HR Manager	2023-05-15 00:00:00	60000
6	NaN	S1002	Jane Doe	Marketing	Marketing Specialist	2023-08-20 00:00:00	50000
7	NaN	S1003	Michael Johnson	Finance	Financial Analyst	2023-10-10 00:00:00	55000
8	NaN	S1004	Emily Brown	Sales	Sales Representative	2023-03-25 00:00:00	48000
9	NaN	S1005	David Wilson	Information Technology	IT Specialist	2023-09-12 00:00:00	60000

```
[38]: import pandas as pd
df=pd.read_json("titanic.json") #Importing json file using pandas
df.head(10) #displays 10 data from the file
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	None	S

```
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```

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JupyterLab Python [conda env:base] * Anaconda Toolbox

```
[38]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	None	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	None	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	None	S
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	None	Q
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7	8	0	3	Paisson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	None	S
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	None	S
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	None	C

```
[39]: print(df.isnull()) #checks for null values - if null(true) not null (false)
```

```
0      PassengerId  Survived  Pclass   Name     Sex    Age  SibSp  Parch  Ticket   \
1      False       False   False   False   False  False  False  False  False
2      False       False   False   False   False  False  False  False  False
3      False       False   False   False   False  False  False  False  False
4      False       False   False   False   False  False  False  False  False
..      ...         ...     ...     ...     ...    ...    ...    ...    ...
886     False       False   False   False   False  False  False  False  False
887     False       False   False   False   False  False  False  False  False
888     False       False   False   False   False  True   False  False  False
889     False       False   False   False   False  False  False  False  False
```

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Python [conda env:base] *

Anaconda Toolbox

```
Fare Cabin Embarked
0 False True False
1 False False False
2 False True False
3 False False False
4 False True False
.. ... ..
886 False True False
887 False False False
888 False True False
889 False False False
890 False True False

[891 rows x 12 columns]

[40]: df.isnull().sum() # total count of null value presents

[40]: PassengerId      0
Survived            0
Pclass             0
Name               0
Sex               0
Age              177
SibSp             0
Parch            0
Ticket            0
Fare              0
Cabin           687
Embarked          2
dtype: int64

[41]: df.tail(10) #displays last 10 data from the dataset
```

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Python [conda env:base] *

Anaconda Toolbox

```
[41]: df.tail(10) #displays last 10 data from the dataset

[41]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
881	882	0	3	Markun, Mr. Johann	male	33.0	0	0	349257	7.8958	None	S
882	883	0	3	Dahlberg, Miss. Gerda Ulrika	female	22.0	0	0	7552	10.5167	None	S
883	884	0	2	Banfield, Mr. Frederick James	male	28.0	0	0	C.A./SOTON 34068	10.5000	None	S
884	885	0	3	Sutehall, Mr. Henry Jr	male	25.0	0	0	SOTON/OQ 392076	7.0500	None	S
885	886	0	3	Rice, Mrs. William (Margaret Norton)	female	39.0	0	5	382652	29.1250	None	Q
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	None	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	None	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	None	Q

```
[42]: print(df.columns) #checks what are all the columns are there

Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
      'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')

[44]: df[df['Age'].isnull()] # columns have null or not

[44]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	None	Q
17	18	1	2	Williams, Mr. Charles Eugene	male	NaN	0	0	244373	13.0000	None	S

```
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[44]: df[df['Age'].isnull()] # columns have null or not

[44]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	None	Q
17	18	1	2	Williams, Mr. Charles Eugene	male	NaN	0	0	244373	13.0000	None	S
19	20	1	3	Masselmani, Mrs. Fatima	female	NaN	0	0	2649	7.2250	None	C
26	27	0	3	Emir, Mr. Farred Chehab	male	NaN	0	0	2631	7.2250	None	C
28	29	1	3	O'Dwyer, Miss. Ellen "Nellie"	female	NaN	0	0	330959	7.8792	None	Q
...	...	...	...	...	...	...	...	...	...	...	...	...
859	860	0	3	Razi, Mr. Raihed	male	NaN	0	0	2629	7.2292	None	C
863	864	0	3	Sage, Miss. Dorothy Edith "Dolly"	female	NaN	8	2	CA. 2343	69.5500	None	S
868	869	0	3	van Melkebeke, Mr. Philemon	male	NaN	0	0	345777	9.5000	None	S
878	879	0	3	Laleff, Mr. Kristo	male	NaN	0	0	349217	7.8958	None	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	None	S

```
177 rows x 12 columns

[45]: #fillna()
#replace missing values
df['Age'].mean()

[45]: 29.69911764705882

[47]: mode_embarked=df['Embarked'].mode()[0]
df['Embarked']=df['Embarked'].fillna(mode_embarked)
```

```
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[61]: import pandas as pd
import json
df = pd.DataFrame(data).set_index('PassengerId') #change my index to PassengerId
print(df)

PassengerId  Survived  Pclass \
1            0         3
2            1         1
3            1         3
4            1         1
5            0         3
...         ...      ...
887          0         2
888          1         1
889          0         3
890          1         1
891          0         3

PassengerId  Name  Sex  Age \
1            Braund, Mr. Owen Harris  male  22.0
2  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0
3            Heikkinen, Miss. Laina  female  26.0
4  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0
5            Allen, Mr. William Henry  male  35.0
...         ...      ...
887          Montvila, Rev. Juozas  male  27.0
888          Graham, Miss. Margaret Edith  female  19.0
889  Johnston, Miss. Catherine Helen "Carrie"  female  NaN
890          Behr, Mr. Karl Howell  male  26.0
891          Dooley, Mr. Patrick  male  32.0

PassengerId  SibSp  Parch  Ticket  Fare  Cabin  Embarked
```

```
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PassengerId  Survived  Pclass  SibSp  Parch  Ticket  Fare  Cabin  Embarked
1            0         3         1     0    A/5 21171   7.2500  None    S
2            1         1         0     0    PC 17599  51.2833   C85    C
3            0         3         0     0  STON/O2. 3101282  7.9250  None    S
4            1         3         1     0   113803  53.1000  C123    S
5            0         3         0     0   373450   8.0500  None    S
...         ...      ...      ...  ...  ...
887          0         2         0     0   211536  13.0000  None    S
888          1         1         0     0   112053  30.0000  B42    S
889          0         3         1     2    W./C. 6607  23.4500  None    S
890          0         3         0     0   111369  30.0000  C148    C
891          0         3         0     0   370376   7.7500  None    Q

[891 rows x 11 columns]

[63]: #groupby
#average fare per class
print(df.groupby('Pclass')['Fare'].mean())

Pclass
1    84.154687
2    20.662183
3    13.675550
Name: Fare, dtype: float64

[66]: #count survivors by gender
df.groupby('Sex')['Survived'].sum()

Sex
female    233
male      109
Name: Survived, dtype: int64

[68]: #multiple aggregation
#agg()
print(df.groupby('Pclass').agg({
```

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```
[68]: #multiple aggregation
#agg()
print(df.groupby('Pclass').agg([
    'Age':['mean','min','max'],
    'Fare':['mean','sum']
]))
```

	Age			Fare	
	mean	min	max	mean	sum
Pclass					
1	38.233441	0.92	80.0	84.154687	18177.4125
2	29.877630	0.67	70.0	20.662183	3801.8417
3	25.140620	0.42	74.0	13.675550	6714.6951

```
[70]: #transform()
#returns same number of rows
#add class average fare
df['Avg_Class_Fare']=df.groupby('Pclass')['Fare'].transform('mean')
print(df)
```

	Survived	Pclass	\
PassengerId			
1	0	3	
2	1	1	
3	1	3	
4	1	1	
5	0	3	
...	...	...	
887	0	2	
888	1	1	
889	0	3	
890	1	1	
891	0	3	

Name Sex Age \

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JupyterLab Python [conda envbase] Anaconda Toolbox

```
4 Futrelle, Mrs. Jacques Heath (Lily May Peel) female 35.0
5 Allen, Mr. William Henry male 35.0
...
887 Montvila, Rev. Juozas male 27.0
888 Graham, Miss. Margaret Edith female 19.0
889 Johnston, Miss. Catherine Helen "Carrie" female NaN
890 Behr, Mr. Karl Howell male 26.0
891 Dooley, Mr. Patrick male 32.0
```

	SibSp	Parch	Ticket	Fare	Cabin	Embarked	\
PassengerId							
1	1	0	A/5 21171	7.2500	None	S	
2	1	0	PC 17599	71.2833	C85	C	
3	0	0	STON/O2. 3101282	7.9250	None	S	
4	1	0	113803	53.1000	C123	S	
5	0	0	373450	8.0500	None	S	
...	...	...	...	...	...	...	
887	0	0	211536	13.0000	None	S	
888	0	0	112053	30.0000	B42	S	
889	1	2	W./C. 6607	23.4500	None	S	
890	0	0	111369	30.0000	C148	C	
891	0	0	370376	7.7500	None	Q	

	Avg_Class_Fare
PassengerId	
1	13.675550
2	84.154687
3	13.675550
4	84.154687
5	13.675550
...	...
887	20.662183
888	84.154687
889	13.675550
890	84.154687
891	13.675550



```
[75]: import pandas as pd
students=pd.read_excel("Merge_join_Concat_Dataset_50_Records.xlsx",sheet_name="students")
marks=pd.read_excel("Merge_join_Concat_Dataset_50_Records.xlsx",sheet_name="marks")
contact=pd.read_excel("Merge_join_Concat_Dataset_50_Records.xlsx",sheet_name="contact")
```

```
#1. inner merge
#returns only matching students_ids from both dataframes.
```

```
inner_df=pd.merge(
    students,
    marks,
    on="Student_ID",
    how="inner"
)
print(inner_df.head())
print(inner_df.shape)
```

	Student_ID	Name	Department	Maths	Python
0	1001	Student_1	CSE	61	56
1	1002	Student_2	IT	62	57
2	1003	Student_3	ECE	63	58
3	1004	Student_4	EEE	64	59
4	1005	Student_5	MECH	65	60

(45, 5)

```
[77]: #LEFT MERGE
#keep all records from the Left Dataframe(students)
```

```
left_df=pd.merge(
    students,
    marks,
    on='Student_ID',
    how="left"
)
```



```
print(left_df.tail())
```

```
#even if marks are missing, students remain
```

	Student_ID	Name	Department	Maths	Python
45	1046	Student_46	CSE	NaN	NaN
46	1047	Student_47	IT	NaN	NaN
47	1048	Student_48	ECE	NaN	NaN
48	1049	Student_49	EEE	NaN	NaN
49	1050	Student_50	MECH	NaN	NaN

```
[78]: #OUTER MERGE
```

```
outer_df=pd.merge(
    students,
    marks,
    on='Student_ID',
    how="outer"
)
print(outer_df.tail())
```

	Student_ID	Name	Department	Maths	Python
50	2001	NaN	NaN	75.0	80.0
51	2002	NaN	NaN	75.0	80.0
52	2003	NaN	NaN	75.0	80.0
53	2004	NaN	NaN	75.0	80.0
54	2005	NaN	NaN	75.0	80.0

```
[82]: #MERGE MULTIPLE TABLES
```

```
#COMBINE students+ marks+contact
```

```
student_complete=pd.merge(
    students,
    marks,
    on='Student_ID',
    how="left"
)
```

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JupyterLabPython [conda env:base] *Anaconda

```
print(student_complete.head())
```

	Student_ID	Name	Department	Maths	Python	Email
0	1001	Student_1	CSE	61.0	56.0	student1@college.edu
1	1002	Student_2	IT	62.0	57.0	student2@college.edu
2	1003	Student_3	ECE	63.0	58.0	student3@college.edu
3	1004	Student_4	EEE	64.0	59.0	student4@college.edu
4	1005	Student_5	MECH	65.0	60.0	student5@college.edu

```
[83]: #Concat row-wise

#split students into two parts

part1=students.iloc[:25]
part2=students.iloc[25:]
combined_row=pd.concat([part1,part2])
print(combined_row.shape)

(50, 3)

[84]: #Concat column-wise
student_basic=students[['Student_ID','Name']]
student_marks=marks[['Python']]
combined_cols=pd.concat(
    [student_basic,student_marks],
    axis=1
)
print(combined_cols.head())
```

	Student_ID	Name	Python
0	1001	Student_1	56
1	1002	Student_2	57
2	1003	Student_3	58
3	1004	Student_4	59
4	1005	Student_5	60

```
[85]: #join()
```

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JupyterLabPython [conda env:base] *Anaconda Toolbox

```
contact_index=contact.set_index("Student_ID")
joined_df=students_index.join(contact_index)
print(joined_df.tail(10))
```

	Student_ID	Name	Department	Email
	1041	Student_41	CSE	student41@college.edu
	1042	Student_42	IT	student42@college.edu
	1043	Student_43	ECE	student43@college.edu
	1044	Student_44	EEE	student44@college.edu
	1045	Student_45	MECH	student45@college.edu
	1046	Student_46	CSE	student46@college.edu
	1047	Student_47	IT	student47@college.edu
	1048	Student_48	ECE	student48@college.edu
	1049	Student_49	EEE	student49@college.edu
	1050	Student_50	MECH	student50@college.edu

Pivot table concept

```
[86]: import pandas as pd
df=pd.read_csv("people-100.csv")
print(df.head())
```

	Index	User Id	First Name	Last Name	Sex	\
0	1	88f7b33d2bcf9f5	Shelby	Terrell	Male	
1	2	f90cd3e76f1a9b9	Phillip	Summers	Female	
2	3	Dbeab8c2dfeFC2c	Kristine	Travis	Male	
3	4	A31Bee3c201ef58	Yesenia	Martinez	Male	
4	5	1ba7A3dc874da3c	Lori	Todd	Male	

	Email	Phone	Date of birth	\
0	elijah57@example.net	001-084-906-7849x73518	1945-10-26	
1	bethany14@example.com	214.112.6044x4913	1910-03-24	
2	bthompson@example.com	277.609.7938	1992-07-02	

```
[87]: print(df.columns)

Index(['Index', 'User Id', 'First Name', 'Last Name', 'Sex', 'Email', 'Phone',
      'Date of birth', 'Job Title'],
      dtype='object')
```

```
[89]: #Create Age Column
df['Date of birth']=pd.to_datetime(df['Date of birth'])
today=pd.Timestamp.today()
df['Age']=(
    (today-df['Date of birth']).dt.days//365
)
print(df[['First Name','Sex','Age']].head())

   First Name  Sex  Age
0     Shelby  Male   80
1    Phillip  Female 116
2   Kristine   Male   33
3    Yesenia   Male    8
4        Lori   Male   87
```

```
[91]: #Basic Pivot Table

#average age by gender

pivot1=pd.pivot_table(
    df,
    values='Age',
    index='Sex'
)
print(pivot1)

           Age
Sex
Female  64.981132
Male    54.957447
```

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```
[96]: #pivot table with aggfunc='mean'

pivot2=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='mean'
)
print(pivot2)
```

```
           Age
Sex
Female  64.981132
Male    54.957447
```

```
[97]: #pivot table with aggfunc='min'

pivot3=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='min'
)
print(pivot3)
```

```
           Age
Sex
Female      8
Male        5
```

```
[98]: #pivot table with aggfunc='max'

pivot4=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='max'
```

```
[99]: #pivot table with aggfunc='sum'
pivot5=pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='sum'
)
print(pivot5)
```

```
      Age
Sex
Female  3444
Male    2583
```

```
[106]: import pandas as pd
df=pd.read_excel("Data_Cleaning_50Plus_Records.xlsx")
print(df.head())

#rename column
df.rename(columns={
    'Student ID' : 'Student_ID',
    'Student Name' : 'Student_Name',
    'Department' : 'Department'
},inplace=True)
print(df.columns)

#3.changing data type
#check data type
print(df.dtypes)
```

```
      Student_ID Student_Name Department Age Salary Email
0           101     student1         IT  20.0  23075.0 student1@gmail.com
1           102     student2         ECE  18.0  35929.0 student2@gmail.com
2           103     STUDENT3         EEE  22.0  29718.0 student3@gmail.com
3           104     student4         MECH  20.0  21009.0 student4@gmail.com
```

```
[108]: #convert Student_ID to String
df['Student_ID']=df['Student_ID'].astype(str)

#convert age to int
#first handle missing values
df['Age']=df['Age'].fillna(0)
df['Age']=df['Age'].astype(int)
#check:
print(df.dtypes)
```

```
Student_ID      object
Student_Name    object
Department      object
Age             int32
Salary          float64
Email           object
dtype: object
```

```
[109]: #Detecting duplicates records
#find duplicates
duplicates=df.duplicated()
print(duplicates)
```

```
0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9    False
10   False
11   False
12   False
```